Computer Science & IT

Theory of Computation

Context Free Languages

Lecture No. 5

By- Ravindra Sir

Topics to be Covered

Topic

Topic

Topic

CFL

CYK

Ambiguous grammars

ambiguous languages

$ww \rightarrow CSL$

$w\overline{w} \rightarrow CSL$

$\overline{ww} \rightarrow CSL$

$a^n b^n c^n \rightarrow CSL \uparrow CSL$

CYK → CNF

Cocke Younger Kasami

CYK is membership algo

$w \overset{?}{\in} L$

If there is atleast one derivation tree $\Longleftrightarrow$ atleast 1 LMD

atleast 1 RMD ✓

CYK, ambiguous

$\underline{\underline{G}}$

$S \to AB / BC$

$A \to BA / a$

$B \to CC / b$

$C \to AB / a$

CYK?

$\underline{\underline{baaba}} \overset{?}{\in} L(G)$

Yes or no

3 min

How many substrings of $\underline{\underline{baaba}} \in L(G)$?

CYK

3 min

$S \rightarrow AB / BC$

$A \rightarrow BA / a$

$B \rightarrow CC / b$

$C \rightarrow AB / a$

↓

CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1					B
2				A,C	
3			A,C		
4		B			
5	A,C				

b a a b a

1 2 3 4 5

$S \to AB / BC$
$A \to BA / a$
$B \to CC / b$
$C \to AB / a$

↓ CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1				A,S	B
2				A,C	
3			A,C		
4		B			
5	A,C				

1 | 2

11 : 22

B : (A,C)

BA, BC

$S \rightarrow AB / BC$

$A \rightarrow BA / a$

$B \rightarrow CC / b$

$C \rightarrow AB / a$

$\downarrow$ CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1				A,S	B
2			B	A,C	
3			A,C		
4		B			
5	A,C				

2 | 3

22 . 33

(A,C) · (A,C)

AA, AC, CA, CC

✗ ✗ ✗ ✓

$S \to AB / BC$

$A \to BA / a$

$B \to CC / b$

$C \to AB / a$

↓

CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1				A,S	B
2			B	A,C	
3		S,C	A,C		
4		B			
5	A,C				

3/4

33 . 44

(A,C) . B

AB ✓ , CB ✗

$S \to AB / BC$
$A \to BA / a$
$B \to CC / b$
$C \to AB / a$

$\downarrow$
CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1				A,S	B
2			B	A,C	
3		S,C	A,C		
4	A,S	B			
5	A,C				

4 | 5
$\downarrow$ $\downarrow$
44 . 55
(B) . (A, C)
BA, BC ✓

$S \to AB / BC$
$A \to BA / a$
$B \to CC / b$
$C \to AB / a$

$\downarrow$
CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1			ϕ	A,S	B
2			B	A,C	
3		S,C	A,C		
4	A,S	B			
5	A,C				

1 3

1 | 2 3

11 . 23
(B) (B)
= BB

1 2 | 3

12 33
(A,S) (A C)
AA, AC, SA, SC
✗ ✗ ✗ ✗

$S \rightarrow AB / BC$
$A \rightarrow BA / a$
$B \rightarrow CC / b$
$C \rightarrow AB / a$

CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1			ϕ	A, S	B
2		B	B	A, C	
3		S, C	A, C		
4	A, S	B			
5	A, C				

2 3 | 4
(2 3) (4 4)
B · B

2 4
$\Downarrow$
2 | 3 4
(2 2) (3 4)
(A C) (S C)
AS AC CS CC
× × × ✓

$S \to AB / BC$
$A \to BA / a$
$B \to CC / b$
$C \to AB / a$

CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1			$\emptyset$	A,S	B
2		B	B	A,C	
3		S,C	A,C		
4	A,S	B			
5	A,C				

2 3 | 4

(2 3) (4 4)

B . B

2 4

$\Downarrow$

2 | 3 4

(2 2) (3 4)

(A C) (S C)

AS AC CS CC

× × × ✓

5 min   3 min time

$S \to AB / BC$

$A \to BA / a$

$B \to CC / b$

$C \to AB / a$

$\downarrow$

CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1			ϕ	A,S	B
2		B	B	A,C	
3		S,C	A,C		
4	A,S	B			
5	A,C				

5 min   3 min time

3 5

$\Downarrow$

3 | 4 5

(3 3) (4 5)

(AC) (AS)

AA, AS, CA, CS

× × ✗ ✗

$S \to AB / BC$
$A \to BA / a$
$B \to CC / b$
$C \to AB / a$

$\downarrow$
CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1			ϕ	A,S	B
2		B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

35
$\Downarrow$
34 | 5

(34) (55)
(S,C) (A,C)

SA, SC, CA, CC
✗ ✗ ✗ ✓

5 min 3 min time

$S \to AB / BC$

$A \to BA / a$

$B \to CC / b$

$C \to AB / a$

$\downarrow$ CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1		✓	$\emptyset$	A,S	B
2		B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

5 min 3 min time

1 4

$\Downarrow$

1 | 2 3 4

11 24

(B) (B)

BB

x

$S \rightarrow AB/BC$
$A \rightarrow BA/a$
$B \rightarrow CC/b$
$C \rightarrow AB/a$

↓
CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1		✓	ϕ	A,S	B
2		B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

1 4
⇓
1 2 | 3 4

(1 2) (3 4)
(A,S) (S,C)
AS, AC, SS, SC
× × × ×

5 min 3 min time

$S \to AB / BC$
$A \to BA / a$
$B \to CC / b$
$C \to AB / a$

↓ CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1		ϕ	ϕ	A,S	B
2	✓	B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

5 min   3 min time

1 4

1 2 3 | 4

1 3   4 4

$\phi \cdot (\ ) = \phi$

$S \rightarrow AB/BC$
$A \rightarrow BA/a$
$B \rightarrow CC/b$
$C \rightarrow AB/a$

$\downarrow$
CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1		ϕ	ϕ	A,S	B
2	S,C	B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

5 min   3 min time

25
$\Downarrow$
2 | 3 4 5
(22) (35)
(A,C) (B)
AB CB

$S \rightarrow AB / BC$

$A \rightarrow BA / a$

$B \rightarrow CC / b$

$C \rightarrow AB / a$

↓ CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1		ϕ	ϕ	A,S	B
2	S,C,A	B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

5 min   3 min time

25

↓ ✓

2 3 | 4 5

(2 3) (4 5)

(B) (A,S)

BA, BS

$S \to AB / BC$

$A \to BA / a$

$B \to CC / b$

$C \to AB / a$

↓

CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1		ϕ	ϕ	A,S	B
2	S,C,A	B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

25

2 3 4 | 5

(24) (55)

B × A,C

BA BC

5 min

3 min time

$S \rightarrow AB / BC$
$A \rightarrow BA / a$
$B \rightarrow CC / b$
$C \rightarrow AB / a$

↓
CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1	S,A	ϕ	ϕ	A,S	B
2	S,C,A	B	B	A,C	
3	B	S,C	A,C		
4	A, S	B			
5	A,C				

15

$1 \mid 2 \quad 3 \quad 4 \quad 5$

(11) (25)
(B) (S,C,A)

BS BC BA

5 min 3 min time

$S \rightarrow AB / BC$

$A \rightarrow BA / a$

$B \rightarrow CC / b$

$C \rightarrow AB / a$

$\downarrow$ CNF

b a a b a ✓

1 2 3 4 5

	5	4	3	2	1
1	S,A,C	ϕ	ϕ	A,S	B
2	S,C,A	B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

5 min   3 min time

15

1 2 | 3 4 5

(1 2) (3 5)

(A,S) (B)

AB, SB

x

$S \rightarrow AB / BC$
$A \rightarrow BA / a$
$B \rightarrow CC / b$
$C \rightarrow AB / a$

$\downarrow$ CNF

b a a b a ✓
1 2 3 4 5

	5	4	3	2	1
1	S,A,C	ϕ	ϕ	A,S	B
2	S,C,A	B	B	A,C	
3	B	S,C	A,C		
4	A,S	B			
5	A,C				

$\underline{5 \text{ min}}$ 3 min time

1 5
1 2 3 | 4 5

1 3 4 5
ϕ $= \phi$

$S \to AB / BC$

$A \to BA / a$

$B \to CC / b$

$C \to AB / a$

↓

CNF

b a a b a

1 2 3 4 5

	5	4	3	2	1
1	S, A, C	ϕ	ϕ	A, S	B
2	S, C, A	B	B	A, C	
3	B	S, C	A, C		
4	A, S	B			
5	A, C				

$w \in L$

How many substrings of baaba belong to L(G)

baaba

ba

aaba

ab

~~ba~~

5 min

3 min time

$S \rightarrow AB$

$A \rightarrow BB/a$

$B \rightarrow AB/b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1				ϕ	A
2				A	
3			B		
4		B			
5	B				

1 | 2

11 22

A A = AA

$S \rightarrow AB$

$A \rightarrow BB / a$

$B \rightarrow AB / b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1				ϕ	A
2			S, B	A	
3			B		
4		B			
5	B				

2 | 3

22 33

A B = AB

$S \to AB$

$A \to BB / a$

$B \to AB / b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1				ϕ	A
2			S, B	A	
3		A	B		
4		B			
5	B				

3 | 4

33 44

B . B = BB

$S \to AB$
$A \to BB/a$
$B \to AB/b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1				ϕ	A
2			S, B	A	
3		A	B		
4	A	B			
5	B				

4 | 5

44 55

B B

$S \rightarrow AB$
$A \rightarrow BB/a$
$B \rightarrow AB/b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1			S,B	∅	A
2			S,B	A	
3		A	B		
4	A	B			
5	B				

1 3
⇓
1 | 2 3

(1 1) (2 3)
A × (S, B)
AS, AB
× ✓

$S \rightarrow AB$
$A \rightarrow BB / a$
$B \rightarrow AB / b$

1 2 3 4 5
a a b b b

	5	4	3	2	1
1			S,B	ϕ	A
2			S,B	A	
3		A	B		
4	A	B			
5	B				

1 3

1 2 | 3

(1 2) (3 3) $= \phi$

ϕ

$S \rightarrow AB$
$A \rightarrow BB / a$
$B \rightarrow AB / b$

1 2 3 4 5
a a b b b

	5	4	3	2	1
1			S,B	ϕ	A
2			S,B	A	
3		A	B		
4	A	B			
5	B				

2 4
2 | 3 4
(22) (34)
A A
X

$S \rightarrow AB$

$A \rightarrow BB / a$

$B \rightarrow AB / b$

	1	2	3	4	5
	a	a	b	b	b

	5	4	3	2	1
1			S,B	ϕ	A
2		A	S,B	A	
3		A	B		
4	A	B			
5	B				

2 4

2 3 | 4

(2 3) (4 4)

(S,B) (B)

SB , BB

✗ ✓

$S \rightarrow AB$
$A \rightarrow BB / a$
$B \rightarrow AB / b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1			S,B	ϕ	A
2		A	S,B	A	
3		A	B		
4	A	B			
5	B				

3 5

3 | 4 5

(33) (45)

$B \cdot A = BA$ ×

$S \rightarrow AB$
$A \rightarrow BB / a$
$B \rightarrow AB / b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1			S,B	ϕ	A
2		A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

3 5

34 | 5

(34) (55)

$A \cdot B = AB$

$S \rightarrow AB$

$A \rightarrow BB / a$

$B \rightarrow AB / b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1			S,B	ϕ	A
2		A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

14

1 | 2 3 4

(11)(24)

(A)(A) = AA ✗

$S \rightarrow AB$
$A \rightarrow BB / a$
$B \rightarrow AB / b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1			S,B	ϕ	A
2		A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

14

1 2 | 3 4

$\underset{\phi}{(1\ 2)}(3\ 4) = \phi$

$S \rightarrow AB$

$A \rightarrow BB/a$

$B \rightarrow AB/b$

1 2 3 4 5

a a b b b

	5	4	3	2	1
1		A	S,B	ϕ	A
2		A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

14

1 2 3 | 4

(13)(44)

(S,B)(B)

SB, BB

✗ ✓

$S \rightarrow AB$

$A \rightarrow BB / a$

$B \rightarrow AB / b$

$\overset{1}{a}\ \overset{2}{a}\ \overset{3}{b}\ \overset{4}{b}\ \overset{5}{b}$

	5	4	3	2	1
1		A	S,B	ϕ	A
2	S,B	A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

2 5

$\Downarrow$

2 | 3 4 5

(22)(35)

(A) (S,B)

AS , AB

×

$S \rightarrow AB$

$A \rightarrow BB \,/\, a$

$B \rightarrow AB \,/\, b$

	1	2	3	4	5
	a	a	b	b	b

	5	4	3	2	1
1		A	S,B	ϕ	A
2	S,B	A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

2 5

$\Downarrow$

2 3 | 4 5

(2 3) (4 5)

(S B) (A)

SA, BA

x x

$S \rightarrow AB$
$A \rightarrow BB / a$
$B \rightarrow AB / b$

1 2 3 4 5
a a b b b

	5	4	3	2	1
1	S B	A	S,B	$\emptyset$	A
2	S,B	A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

1 5

1 | 2 3 4 5

(11) (25)
A (S,B)

AS, AB
× ✓

$S \to AB$
$A \to BB / a$
$B \to AB / b$

1 2 3 4 5
a a b b b

	5	4	3	2	1
1	SB	A	S,B	ϕ	A
2	S,B	A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

1 5

1 2 | 3 4 5

(1 2)(3 5)

$\phi \quad = \phi$

$S \to AB$

$A \to BB / a$

$B \to AB / b$

1 2 3 4 5

a a b b b

	5	4	3	2	1
1	S B	A	S,B	ϕ	A
2	S,B	A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

1 5

1 2 3 | 4 5

(1 3) (4,5)

(S,B) (A)

SA, BA

X X

$S \rightarrow AB$

$A \rightarrow BB/a$

$B \rightarrow AB/b$

1 2 3 4 5

a a b b b

	5	4	3	2	1
1	S B	A	S,B	ϕ	A
2	S,B	A	S,B	A	
3	S,B	A	B		
4	A	B			
5	B				

1 5

1 2 3 4 | 5

(14) (55)

A B = AB

abbb ✓

$S \to AB$

$A \to BB / a$

$B \to AB / b$

Time complexity

$\underline{O(n^2)}$ cells $\times$ $\underline{O(n)}$

$= O(n^3)$ ✓

Space complexity

$1 + 2 + 3 + \cdots n$

$\frac{n(n+1)}{2} = O(n^2)$

	1	2	3	4	5
	a	a	b	b	b

	5	4	3	2	1
1	S, B	A	S, B	∅	A
2	S, B	A	S, B	A	
3	S, B	A	B		
4	A	B			
5	B				

$aabbb \in L(G)$

How many substrings of aabbb are in the language

$\underbrace{1 \mid 2 \mid 3 \mid 4 \mid 5}$

aabbb ✓

aab ✓

abbb ✓

ab ✓

bbb ✓

THANK - YOU

WW → CSL ⇒ CSL ✓